01 /04

Lab Component Sorter: ESP32 + Image Recognition over Wi-Fi

TYPE
Group project, team of 5

DATE
Winter 2025

ROLE
Firmware and pin mapping

STACK

ESP32-CAM | Arduino Nano | Servos | Image recognition | Wi-Fi | 3D printing

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FIG. 1 — CAD: CHUTE + BINS

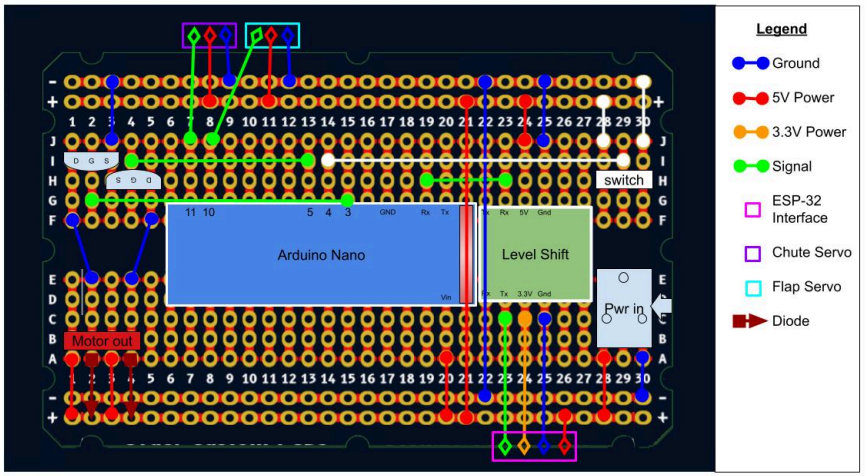


FIG. 2 — PROTOTYPE BOARD LAYOUT

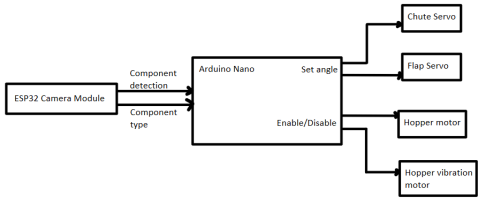


FIG. 3 — SYSTEM DIAGRAM

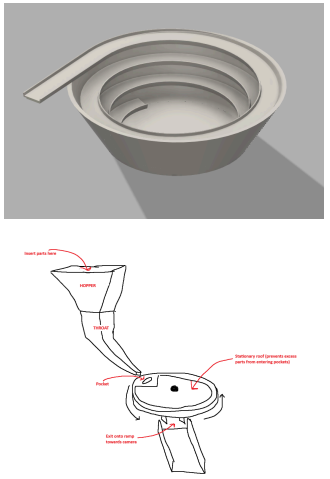


FIG. 4 — ISOLATOR BOWL

02/04

Optics Lab: Interferometry and Data Analysis in MATLAB

TYPE

Carleton physics lab

ROLE

Setup and analysis, with a partner

STACK

MATLAB | Least-squares fitting | Error propagation | Formal reporting

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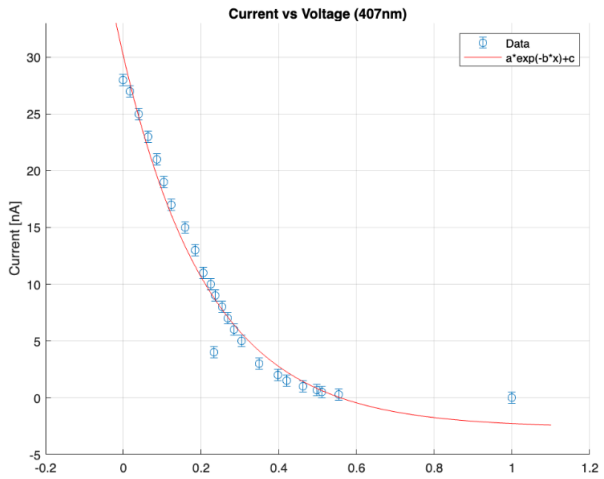


FIG. 1 — NONLINEAR FIT, MATLAB

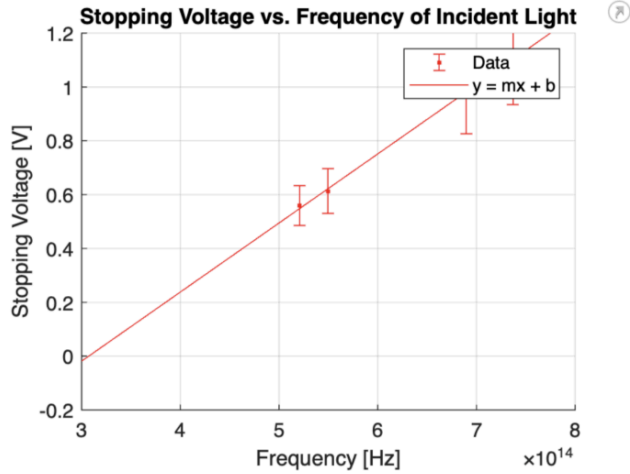


FIG. 2 — LINEAR FIT WITH ERROR BARS

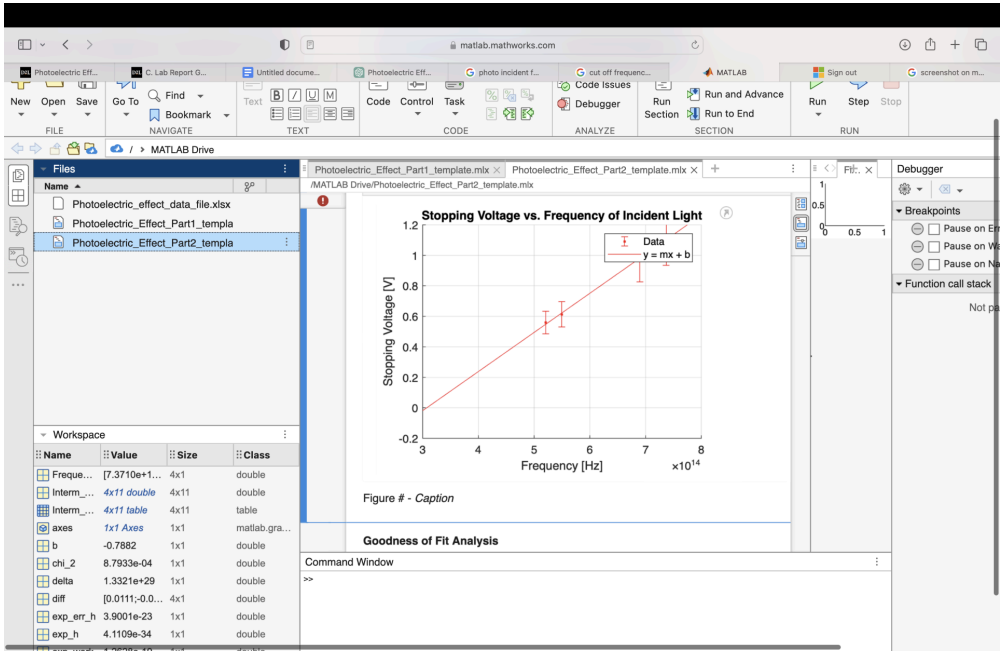


FIG. 3 — MATLAB LIVE SCRIPT AND WORKSPACE

03/04

ESP32 Sensor Board: KiCad PCB + Firmware

TYPE
Personal project

ROLE
Solo: schematic, layout, firmware

RESULT
From-scratch FFT matches NumPy to 2e-6

STACK

KiCad 10 | ESP32 | MPU6050 | SPH0645 | I2C | I2S | C / Arduino

REPO
/pcb-sensor-array

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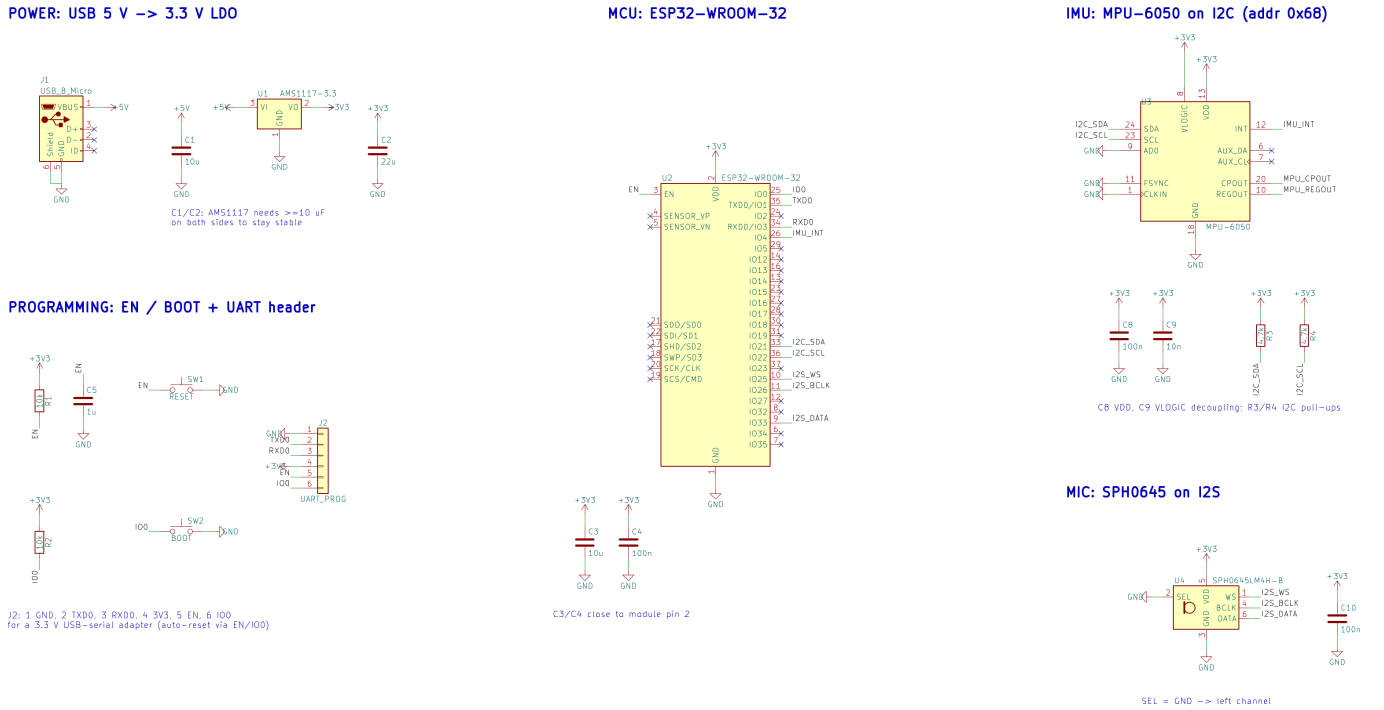


FIG. 1 — KICAD SCHEMATIC

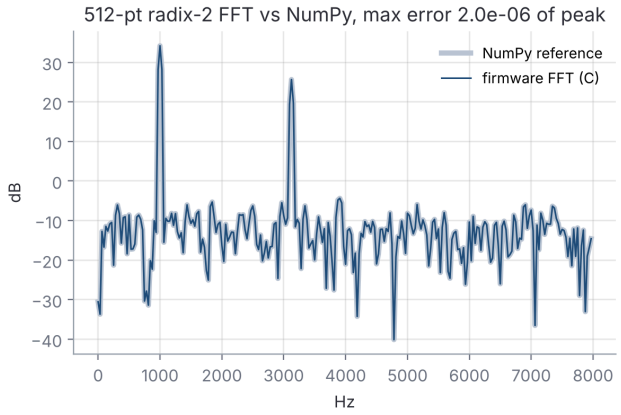


FIG. 2 — FIRMWARE FFT VS NUMPY

```
// in-place radix-2 butterflies, sensor_board.ino
for (int len = 2; len <= n; len <<= 1) {
  float ang = -2 * PI / len;
  float wr = cosf(ang), wi = sinf(ang);
  for (int i = 0; i < n; i += len) {
    float cr = 1, ci = 0;
    for (int k = 0; k < len / 2; k++) {
      int a = i + k, b = a + len / 2;
      float tr = xr[b]*cr - xi[b]*ci;
      float ti = xr[b]*ci + xi[b]*cr;
      xr[b] = xr[a] - tr;  xi[b] = xi[a] - ti;
      xr[a] += tr;         xi[a] += ti;
      float ncr = cr*wr - ci*wi;
      ci = cr*wi + ci*wr;  cr = ncr;
    }
  }
}
```

FIG. 3 — FFT CORE, C ON THE ESP32

04 /04

Exploring Radio with an RTL-SDR

TYPE
Personal project

ROLE
Solo: receiver chain and spectrum tools

RESULT
FM: IQ in, 48 kHz audio out. FFT: peaks within 0.19 Hz, 50 Hz–16 kHz

STACK
RTL-SDR · Python · NumPy · SciPy · FIR design · FM demod · FFT

REPOS
/fm-receiver
/audio-fft-analyzer

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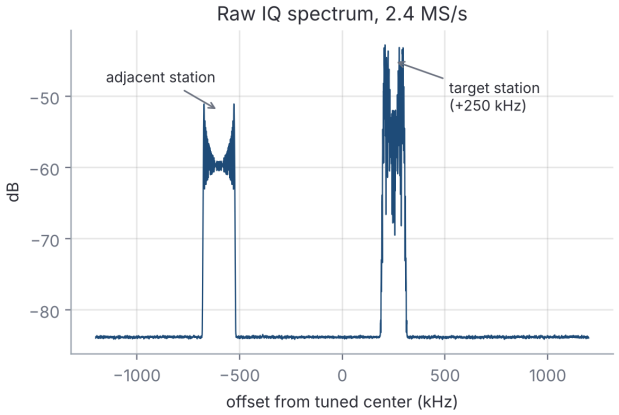


FIG. 1 — IQ SPECTRUM, TWO FM STATIONS

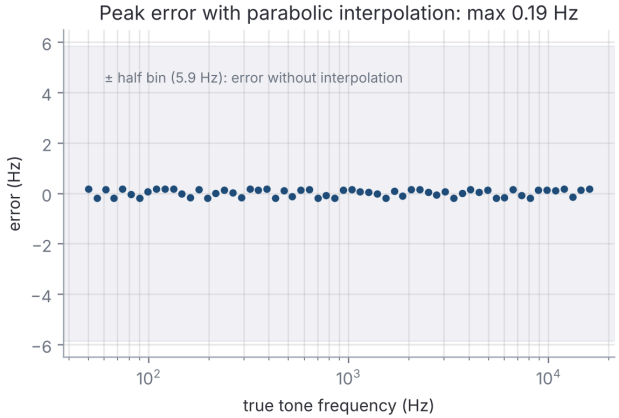


FIG. 2 — FFT PEAK ACCURACY TEST

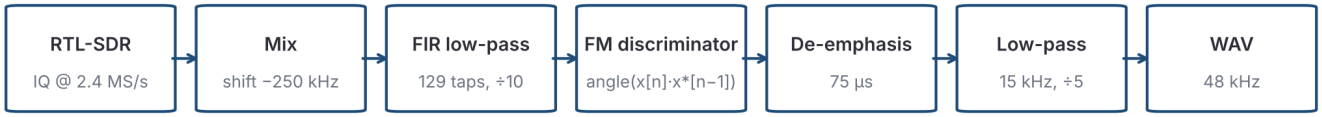


FIG. 3 — FM RECEIVER SIGNAL CHAIN